AI in Science: Case Studies

D-Lab AI Pulse — Session 6

March 31, 2026

Previous Sessions → This Session

Previous sessions demoed specific tools:

- Claude Code, Gemini CLI, Perplexity, Consensus, Elicit, NotebookLM...
- "Here's a tool, here's how to use it"

This session is different — not about any single tool. Instead:

- **Workflows:** How AI fits into end-to-end research pipelines — from problem setup through verification. Each case shows the complete chain of decisions a researcher made.
- **Combination of tools:** Human expertise + AI in iterative loops — prompting, reviewing, feeding results back, refining. The human stays in the loop at every stage.
- **Customization:** From off-the-shelf prompting to fine-tuning domain-specific models on your own data — and when each level of investment pays off.

The Big Players (What We're NOT Covering)

- **AlphaFold** (DeepMind) — Predicted the 3D structure of nearly every known protein. Nobel Prize in Chemistry 2024.
- **GNoME** (DeepMind) — Discovered 2.2 million new stable crystal structures for materials science.
- **GraphCast / GenCast** (DeepMind) — Weather forecasting that beats traditional numerical models. Probabilistic forecasts 15 days out.
- **AlphaGeometry 2** (DeepMind) — Solves olympiad-level geometry problems, approaching gold-medalist performance.

These are **billion-dollar teams** with custom architectures and massive compute. Important — but not what most researchers can do.

The Six Cases

#	Field	What the AI Did
1	Astronomy	Classified cosmic events from 15 examples
2	Biology	Hypothesized cancer drug combinations
3	Social Science	Predicted life outcomes from childhood essays
4	Chemistry	Optimized a novel chemical reaction
5	Theoretical Physics	Simplified & proved a 40-year conjecture
6	Mathematics	Formalized theorems in Lean

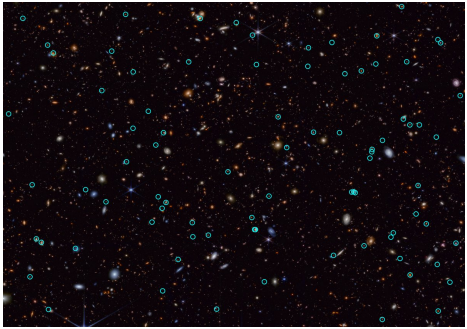
Case 1: Astronomy

Classifying cosmic events from 15 examples

Thousands of transient candidates per night, most are noise. Traditional classifiers need thousands of labels and can't explain their decisions.

Can a general-purpose LLM classify astronomical images with just 15 annotated examples and plain English?

Stoppa, F., Bulmus, T., & Smartt, S. J. (2025).
Textual interpretation of transient image classifications from large language models.
Nature Astronomy.



JADES Transient Survey (NASA/ESA/Webb)

Case 1: Astronomy — The Problem

Drowning in data

- Modern sky surveys: thousands of transient candidates per night
- Most are **noise** — cosmic rays, diffraction spikes, satellite trails, CCD artifacts
- **Coming soon:** Vera Rubin Observatory — ∼10 million alerts/night, 20 TB/day

Traditional ML approach:

- **Convolutional Neural Networks (CNNs):** learn patterns from thousands of labeled training images — typically 5,000–50,000 examples
- **Opaque:** outputs a probability score (e.g., 0.87 "real") with no explanation of *why* — the model can't tell you what features it detected
- **Telescope-specific:** each instrument has different pixel scales, optics, and artifact patterns, so a model trained on one telescope doesn't transfer — you retrain from scratch each time

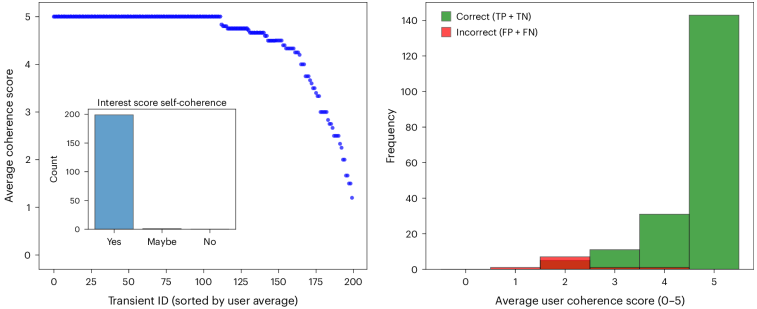
Case 1: Astronomy — The Workflow

15 examples + plain English

1. **Input:** Image triplets (new image, reference image, difference image) for each candidate
2. **Examples:** 15 annotated cases per survey — 9 noise + 6 real, each with a plain-English explanation of *why*
3. **Persona prompt:** *"You are an experienced astrophysicist. . . "*
4. **Run Gemini 1.5 Pro** on all candidates → get classification + natural language explanation + interest score (JSON)
5. **Self-review:** Gemini rates its own explanation coherence (0–5 scale)
6. **Iterate:** Add low-coherence cases to examples, rerun → improved accuracy

Case 1: Astronomy — The Self-Correction Loop

1. **First pass:** 93.4% accuracy
2. **Self-review:** Gemini judges its own explanations on a 0–5 coherence scale
3. **Key finding:** Low coherence ↔ misclassification — the model knows when it's uncertain
4. **Add challenging cases** → rerun → **96.7%**



Stoppa et al. (2025), Fig. 3 — Coherence scores from 12 astronomers

Case 1: Astronomy — The Takeaway

Does **NOT** beat best CNNs (99.5%) — but:

- No training pipeline, no GPUs, no ML expertise
- Works across telescopes without retraining
- Every classification comes with an **explanation**

What went wrong: Too slow and expensive for 10 million alerts/night.

Astronomy — The Few-Shot Recipe

1. Define your classification task
2. Curate ∼15 labeled examples spanning easy and hard cases
3. Write a plain-English explanation for each example
4. Set a persona and output schema in your prompt
5. Run first pass
6. Use self-review to flag low-confidence cases
7. Add hard cases to examples and rerun

This workflow generalizes to any classification task where you have domain expertise and a handful of labeled examples.

Case 2: Biology

Hypothesizing cancer drug combinations

$\sim$1,000 FDA-approved drugs means 499,500 possible pairs. Traditional screening takes 10–15 years with $\sim$0.1% success rate.

Can an LLM narrow half a million possibilities to a handful worth testing — and get it right?

Abdel-Rehim, S., King, R. D., et al. (2025).
Scientific hypothesis generation by large language models: laboratory validation in breast cancer treatment.
J. Royal Society Interface.

Photo: Unsplash

Case 2: Biology — The Problem

499,500 possible drug pairs

- $\sim$1,000 FDA-approved drugs $\rightarrow$ 499,500 possible pairwise combinations
- Traditional drug screening: 10–15 years, $\sim$0.1% success rate
- Breast cancer drug resistance: single agents fail, synergistic combos needed
- *"Every cancer patient ideally requires a scientific research project"*

The question: Can an LLM narrow 499,500 possibilities to a handful worth testing?

Case 2: Biology — The Workflow

Closed-loop AI + automated lab

1. **Define constraints:** FDA-approved, affordable, **non-cancer** drugs that kill MCF7 breast cancer cells but spare healthy MCF10A cells
2. **Prompt GPT-4:** generate 12 drug pair hypotheses with mechanistic reasoning for each
3. **Send to lab:** Arctoris Ltd (automated laboratory, Oxford) tests all 12 combinations
4. **Feed results back:** experimental data returned to GPT-4
5. **Refine:** GPT-4 generates 4 refined combinations based on what worked
6. **Test again:** 3 of 4 show positive synergy — all combinations were novel

Case 2: Biology — "Hallucinations as Hypotheses"

GPT-4's Reasoning

Itraconazole (antifungal)

↓

"Disrupts cell membrane integrity"

↓

Ergosterol synthesis pathway

✗ This mechanism doesn't exist in mammalian cells

Lab Result

Itraconazole + atenolol

Highest synergy score: 4.83
Highest specificity: 7.03

✓ Best-performing combination

"In science some hallucinations may be useful: novel hypotheses whose validity may be tested by laboratory experiments."

Star combo: **disulfiram** (alcoholism) + **simvastatin** (cholesterol) — synergy score **10.58**

Case 2: Biology — The Takeaway

Ross King: creator of Robot Scientist Adam (2009) — first machine to autonomously discover new knowledge

This paper: replacing classical AI hypothesis engine with an LLM

"I don't think anyone would have found them apart from randomly trying things." — King

Vision: *"The AI scientist is no longer a metaphor without experimental validation"*

What went wrong: GPT-4 forgot its own previous recommendations between rounds.

Biology — The Closed-Loop Hypothesis Recipe

1. Define the search space and constraints in natural language
2. Prompt the LLM to generate hypotheses with mechanistic reasoning
3. Test top candidates experimentally
4. Feed results back to the LLM
5. Generate refined hypotheses
6. Test again

Applicable wherever you need novel hypotheses from a large combinatorial space and have a way to test them.

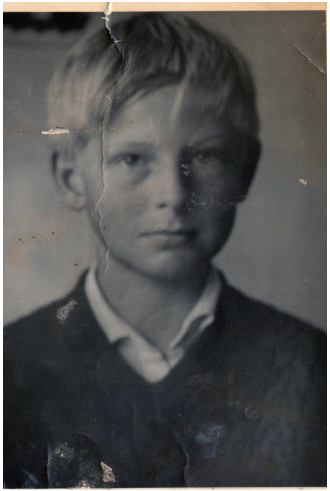
Case 3: Social Science

Predicting life outcomes from childhood essays

17,415 children tracked from birth. At age 11, each wrote a 250-word essay imagining life at 25. Previous attempts to predict life outcomes from childhood data hit a ceiling — 160 research teams couldn't beat $R^2 \approx 0.20$.

Can an LLM embedding of a child's essay outpredict DNA and teacher assessments?

Wolfram, T. (2025).
Large language models predict cognition and education close to or better than genomics or expert assessment.
Communications Psychology / Nature.



NCDS participant age 10 (UCL/CLS)

Case 3: Social Science — The Problem

Can you predict a life from 250 words?

- **National Child Development Study:** every baby born in one week in March 1958 (17,415 children)
- At age 11 (1969): *"Imagine you are now 25 years old. Write about the life you are leading."*
- ∼10,500 essays, averaging 250 words, written in 30 minutes

The "gloomy prospect":

- Fragile Families Challenge (160 teams, 15 years of data): best $R^2 \approx 0.20$
- DNA (polygenic scores): explains only 12–16% of variance

Case 3: Social Science — The Workflow

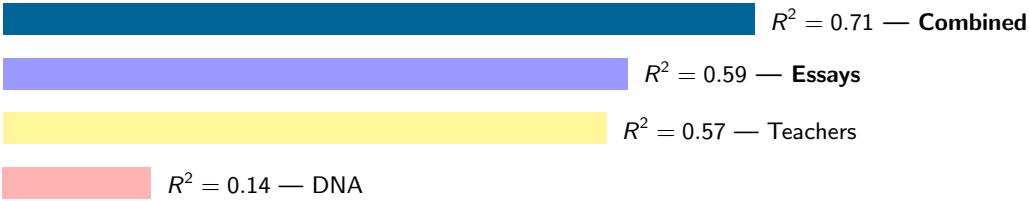
A solo PhD student with an API

1. **Digitize** ∼10,500 childhood essays (250 words each)
2. **Embed:** Send each essay to OpenAI's embedding API → 1,536-dimensional numeric vector capturing meaning, style, and conceptual sophistication
3. **Combine:** Add 534 traditional linguistic metrics (lexical diversity, readability, sentiment, grammatical errors)
4. **Train:** ML ensemble with nested cross-validation
5. **Predict:** Cognitive ability (age 11 & 16), personality (age 50), educational attainment (age 33)

Cost: a few dollars in API calls. **Team:** Tobias Wolfram, Bielefeld University — one person.

Case 3: Social Science — The Results

Reading ability at age 11 — who predicts it best?



Essays beat teachers who knew these children for years — from 250 words in 30 minutes

87% of top features were embedding dimensions — the API call subsumes decades of linguistic feature engineering

Case 3: Social Science — The Takeaway

One researcher, one API, a few dollars → outpredicts billions of dollars of genomics research

Challenges the "gloomy prospect" that life outcomes are unpredictable

"I would not be surprised if prompting an LLM using a chat interface without giving it any training data at all outperforms the results in the paper." — Wolfram

Ethical tension: Wolfram co-founded Herasight, an embryo screening startup claiming to predict intelligence from DNA. The predictive power demonstrated here makes the ethical stakes concrete, not hypothetical.

What went wrong: Raises surveillance and screening concerns the author hasn't resolved.

Social Science — The Embedding Pipeline Recipe

1. Collect your text data
2. Send to an embedding API to get numeric vectors
3. Optionally combine with traditional domain-specific features
4. Train an ML model with cross-validation
5. Interpret which feature set drives prediction

Works for any domain with text data you want to use as a predictor — surveys, clinical notes, policy documents, student writing.

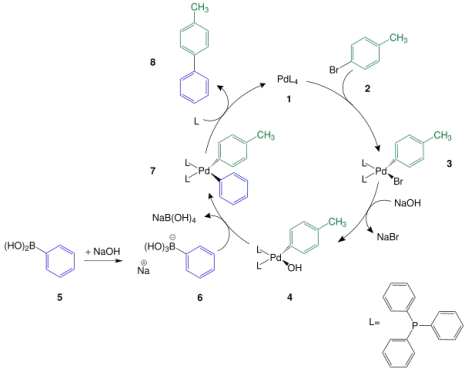
Case 4: Chemistry

Optimizing a novel chemical reaction

Retrosynthesis — working backward from a target molecule to find starting materials — has been a goal for AI since 1969. Optimizing a new reaction means choosing among dozens of catalysts, ligands, solvents, and temperatures: hundreds of trials over weeks.

Can a small, fine-tuned model beat GPT-4 at chemistry?

Zhang, X., Xu, Y., et al. (2025).
Large language models to accelerate organic chemistry synthesis.
arXiv:2504.18340.



Suzuki cross-coupling mechanism (Wikimedia, CC)

Case 4: Chemistry — The Problem

Retrosynthesis since 1969

- **Retrosynthesis:** working backward from a target molecule to find starting materials (Corey, Nobel 1990)
- First computer-aided synthesis program: **1969**. 55+ years of trying.
- Optimizing a new reaction: dozens of catalysts, ligands, solvents, temperatures $\rightarrow$ **hundreds of trials** over weeks
- Chemical space: $\sim 10^{60}$ possible molecules

Case 4: Chemistry — The Fine-Tuning

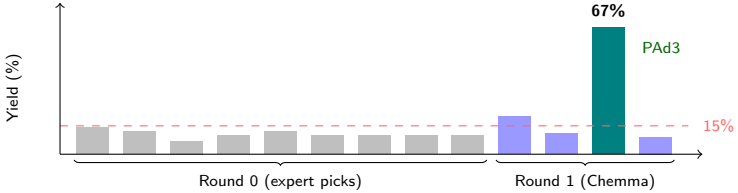
Prompting = use the model as-is (Cases 1–3). **Fine-tuning** = reshape the model for your domain (Case 4).

- Base model: **LLaMA-2-7B** (Meta, open-source, 7 billion parameters)
- Training data: **1.28 million chemistry Q&A pairs** (Open Reaction Database + USPTO patents)
- Hardware: **8 × A800 GPUs, 72 hours** — university lab scale, ∼$2,000–5,000
- Inference: single consumer GPU (∼13 GB) or quantized to ∼4 GB
- Molecules encoded as **SMILES text strings** → chemistry becomes a language task
- **A fine-tuned 7B model beats GPT-4 at chemistry**

Case 4: Chemistry — The Workflow

Target: Previously unreported Suzuki-Miyaura cross-coupling

1. **Round 0 (Runs 1–9):** Test 9 expert-chosen ligands → all yields below 15%
2. **Fine-tune Chemma** on the failure data; chemist switches solvent based on domain judgment
3. **Round 1 (Runs 10–13):** Chemma recommends PAd3 ligand → **67% isolated yield**
4. **Round 2 (Runs 14–15):** Validation and scope expansion



Traditional approach: hundreds of trials over weeks → here: **15 experiments**

Case 4: Chemistry — The Takeaway

Retrosynthesis benchmark: 72.2% accuracy vs. 57.7% previous SOTA (14.5pp jump)

Key insight: Small, specialized, fine-tuned model > large general-purpose model

"It is important that we start to teach students how to effectively use tools like these, as well as how to be critical of their outputs." — Schrier, Fordham

What went wrong: Round 0 completely failed — all expert-chosen ligands gave <15% yield.

Chemistry — The Fine-Tuning Recipe

1. Assemble domain-specific training data as Q&A pairs
2. Pick an open-source base model
3. Fine-tune on your data
4. Deploy for iterative prediction
5. Human expert validates and adjusts experimental conditions each round

Same architecture people run locally with Ollama — but reshaped for your domain with your data.

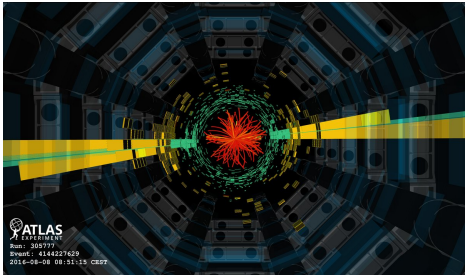
Case 5: Theoretical Physics

Simplifying & proving a 40-year conjecture

Gluon scattering amplitudes were believed to be zero for ∼40 years — stated in textbooks. Physicists found a loophole, derived explicit expressions for small cases, but the algebra exploded in complexity.

Can an LLM spot the pattern hiding in half a page of algebra?

Guevara, A., Lupsasca, A., Skinner, D., & Strominger, A. (2026).
Single-minus gluon tree amplitudes are nonzero.
arXiv:2602.12176.



ATLAS dijet event, 8.12 TeV (CERN, CC-BY-4.0)

Case 5: Theoretical Physics — The Problem

"A textbook result everyone believed for 40 years turned out to have a gap."

- **Gluons:** particles carrying the strong nuclear force (holds quarks together)
- "Single-minus" gluon amplitudes were believed to be **zero since the mid-1980s** — stated in textbooks
- The standard proof relies on choosing a particular reference frame
- **The loophole:** in certain configurations, that choice creates $0/0$ singularities — the proof has a gap

Case 5: Theoretical Physics — The Exploding Complexity

"The math gets exponentially harder — but when simplified, a pattern emerges."

Gluons (n)	**Terms**
3	1
4	2
5	8
6	32 (half a page of algebra)

After simplifying, the 32 terms **collapsed** into a product of 4 simple factors.

GPT-5.2 Pro was fed the $n = 3$ through $n = 6$ simplified expressions.

Think of it as: guess the next number in the sequence — but each number is half a page of algebra.

Case 5: Theoretical Physics — The Workflow

"The AI spotted the pattern and called it 'obvious.'"

1. **Derive** explicit gluon amplitude expressions for $n = 3$ through $n = 6$ by hand
2. **Simplify** each expression by restricting to a special kinematic region
3. **Feed** the simplified sequence to GPT-5.2 Pro
4. **GPT-5.2 conjectures** the general formula: a product of $n - 2$ factors
5. **Separate OpenAI model** writes a complete mathematical proof ($\sim$12 hours)
6. **Human physicists verify** using multiple independent consistency checks

OpenAI Blog — Science Magazine — Harvard Gazette

Case 5: Theoretical Physics — The Takeaway

"Humans set up the problem, AI found the structure, humans verified it."

- The formula was *"not evident from direct inspection"*
- The **authorship question:** GPT-5.2 is not listed as co-author, but "on behalf of OpenAI" appears in the byline
- Strominger (Harvard) visited OpenAI to complete the project — new model of physics collaboration

What went wrong: The AI's proof took 12 hours and required human verification of every step.

Physics — The Pattern-Spotting Recipe

1. Compute explicit results for small cases by hand
2. Simplify each result to reveal structure
3. Feed the simplified sequence to an LLM
4. Ask it to guess the general pattern
5. Verify the conjecture with independent mathematical methods

Works whenever you have a sequence of increasingly complex explicit results and suspect a hidden closed-form pattern.

Case 6: Mathematics

Formalizing theorems in Lean

A mathematical proof convinces a human. A Lean proof is machine-checked — every step explicit and verified. But formalizing is painfully slow: "each line takes about an hour." Finding the right lemma in a library of 100K+ results is a skill unto itself.

Can an AI agent handle the mechanical parts while the mathematician focuses on the ideas?

```
theorem add_comm (n m : Nat)
    : n + m = m + n := by
  induction n with
  | zero => simp
  | succ n ih =>
    simp [Nat.succ_add]
    exact ih
-- No goals
```

✓ *Type-checked — proof verified*

Tao, T. (2026).
Lean Formalization with Claude Code.
YouTube (March 2026).

Case 6: Mathematics — The Problem

Formalizing proofs in Lean 4

A mathematical proof convinces a *human*. A Lean proof is **machine-checked** — every step explicit and verified by the type-checker.

- The gap: *"Each line of proof takes about an hour to formalize"* — Tao, 2023
- Lean's Mathlib library has **100K+ results** — finding the right lemma is a skill unto itself
- Lean's tactic system and error messages add further complexity

Case 6: Mathematics — Three Attempts

Attempt 1 — "Just do the whole thing"

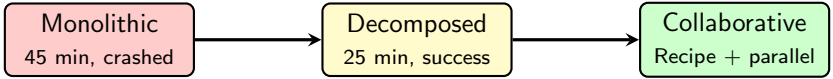
- Monolithic request. Ran 45 minutes, crashed. Produced nothing.

Attempt 2 — Decomposed steps

- Broke proof into subtasks. Succeeded in 25 minutes.

Attempt 3 — Pre-written recipe

- Wrote a blueprint specifying which parts Claude handles and which require manual input.
- Tao and Claude worked **in parallel**. Most effective.



Case 6: Mathematics — What Worked and What Didn't

AI good at: high-level formalization, structuring proofs, generating lemma skeletons

AI bad at: lowest-level mechanical proof steps (counterintuitive!)

"The recipe is where the actual value is" — human planning is the bottleneck

Evolution of workflow:

Case 6: Mathematics — The Takeaway

- **Lean as ground truth:** AI code can be inelegant, but if it type-checks, it's correct
- The world's most famous mathematician showing a **collaborative** workflow — not replacement

"The errors often really subtle and then when you spot them, they're really stupid. No human would have actually made that mistake." — Tao

What went wrong: First attempt ran 45 minutes and crashed with no output.

Mathematics — The Collaborative Blueprint Recipe

1. Write a blueprint specifying which subtasks are human vs AI
2. Decompose the full problem into independent pieces
3. Run the AI on each piece
4. Human reviews and corrects output
5. Iterate until done

Applies to any complex formal task where decomposition is possible and verification is cheap — proofs, code generation, regulatory filings.

"You do need to keep doing that.

Otherwise, if you rely too much on these tools
and something goes wrong,
you may have no idea what to do."

— Terry Tao, March 2026

A Spectrum of Approaches

Prompting
Talk to the model
Cases 2, 5, 6

→

Few-Shot
Show it a handful of examples
Case 1

→

Embeddings
Turn text into numbers
Case 3

→

Fine-Tuning
Retrain the model on your data
Case 4

⟶ increasing technical investment & domain customization

The Anatomy of a Scientific Prompt — Beyond "Chatting"

Casual use: *"Hey, is this image a supernova?"*

Scientific prompting: a structured document with distinct components

Persona / Role: *"You are an experienced astrophysicist. . . "* (Case 1)
Sets the model's frame of reference and domain vocabulary

Task & Constraints: *"FDA-approved, affordable, non-cancer drugs"* (Case 2)
Narrows the search space and prevents drift

Output Schema: Force structured output (JSON, CSV, labeled fields)
Makes results machine-readable and pipeline-ready

Every case in this deck used at least two of these three components.

The Anatomy of a Scientific Prompt — Ensuring Reproducibility

The same prompt can give different answers on different runs — a problem for science.

Three controls:

Temperature = 0 (or near-zero): Makes output near-deterministic — same input, same output

System prompts: Lock in rules that persist across turns — the model can't "forget" its constraints

Few-shot examples: Stabilize formatting and calibrate judgment — show the model what a correct output looks like (Case 1 used 15)

The goal: another researcher with the same prompt, same model, same data should get the same result.

The difference between "I asked ChatGPT" and "we used GPT-4o (temperature=0) with the following system prompt (Appendix A)."

Six Recipes at a Glance

#	Field	Approach	Core Idea	Key Limitation
1	Astronomy	Few-Shot	15 labeled examples + self-review loop	Too slow for 10M alerts/night
2	Biology	Prompting	Hypothesize → test → feed back → refine	LLM forgets between rounds
3	Social Sci.	Embeddings	Text → numeric vectors → ML prediction	Surveillance/screening concerns
4	Chemistry	Fine-Tuning	Reshape a 7B model with 1.28M Q&A pairs	Round 0 failed completely
5	Physics	Prompting	Simplified small cases → ask for pattern	12h proof + human verification
6	Mathematics	Prompting	Blueprint + decompose + iterate	First attempt crashed

Patterns Across the Six Cases

The human stays in the loop:

- Stoppa selected the 15 examples. King designed the prompts. Wolfram built the pipeline. Zhang's team ran the experiments. Physicists set up the problem. Tao wrote the recipe.

AI as different things:

- Classifier (astronomy), hypothesis generator (cancer), reader (essays), domain expert (Chemma), pattern-spotter (gluons), brainstorming partner (Tao)

Failure is part of the process:

- Every case had a failure or limitation that shaped the final approach

What Would It Take to Do This in YOUR Field?

Approach	What You Need	Cost	Skill Level
Prompting (Cases 2, 5, 6)	API access + domain expertise	$20/month subscription	Low
Few-shot (Case 1)	∼15 labeled examples + clear instructions	API costs	Low–Medium
Embeddings (Case 3)	Text data + ML pipeline	API + compute	Medium
Fine-tuning (Case 4)	Training data + GPU access (e.g., Savio)	$2K–5K compute	Medium–High

The common thread: domain expertise matters more than technical AI skill

Resources & Links

Papers:

- Astronomy: Nature Astronomy (Oct 2025)
- Cancer drugs: J. Royal Society Interface (June 2025)
- Essays: Communications Psychology (July 2025)
- Chemma: arXiv:2504.18340 (April 2025)
- Gluons: arXiv:2602.12176 (Feb 2026) — OpenAI Blog
- Tao: Lean Formalization Video (March 2026)

Code & Demos:

- github.com/turanbulmus/spacehack (Astronomy)
- github.com/tobiaswolfram/llm_paper (Essays)
- huggingface.co/AI4Chem — ai4chem.sjtu.edu.cn (Chemma)

Discussion

Questions, reactions, ideas for your own field?